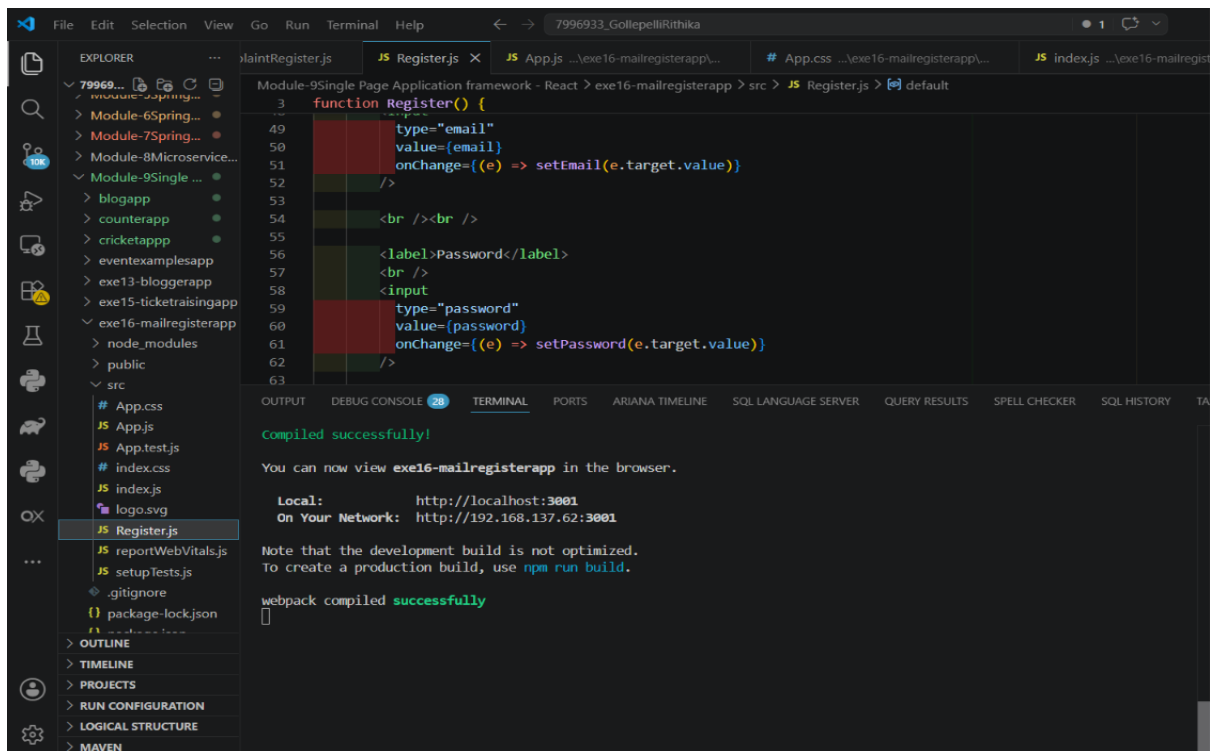


Output16exe16-mailregisterapp.ReactJS-HOL



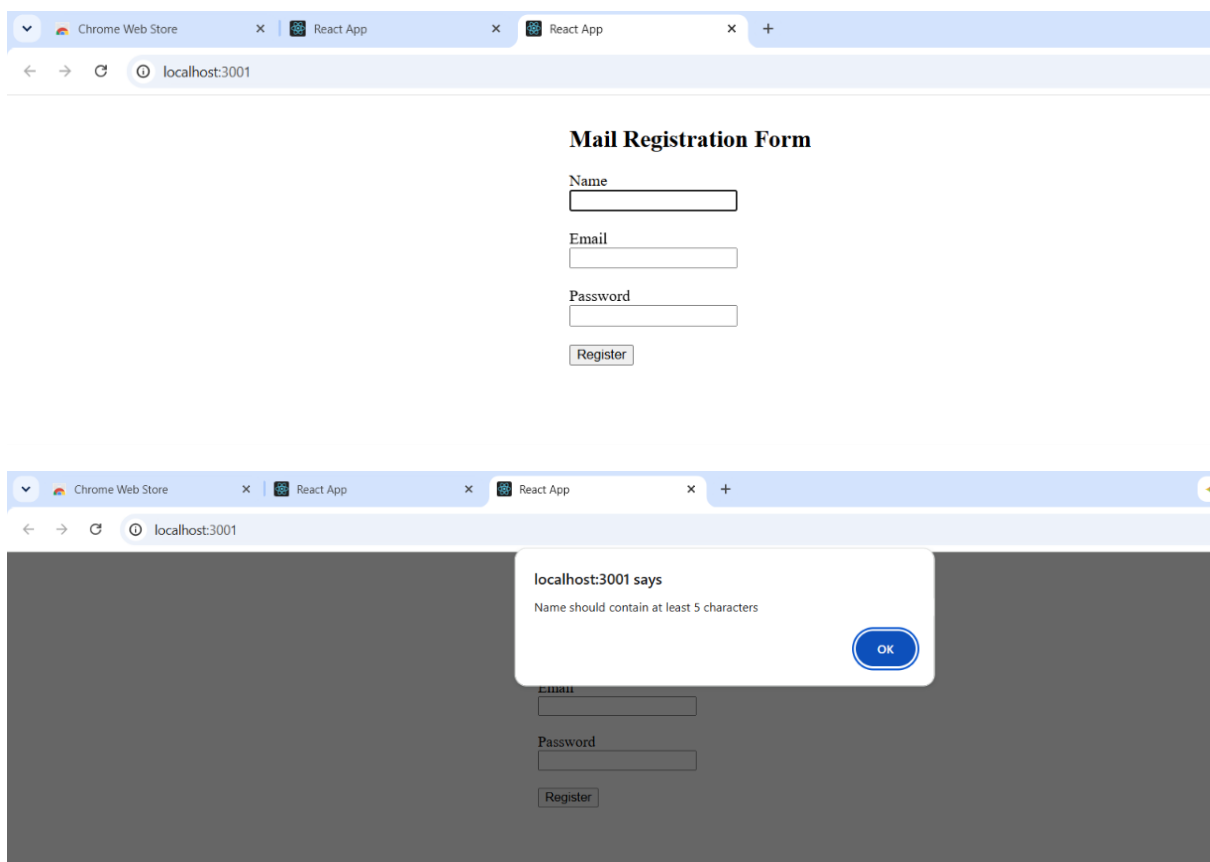
The screenshot shows the Visual Studio Code editor with the file explorer on the left displaying the project structure. The main editor window shows the `Register.js` file with the following code:

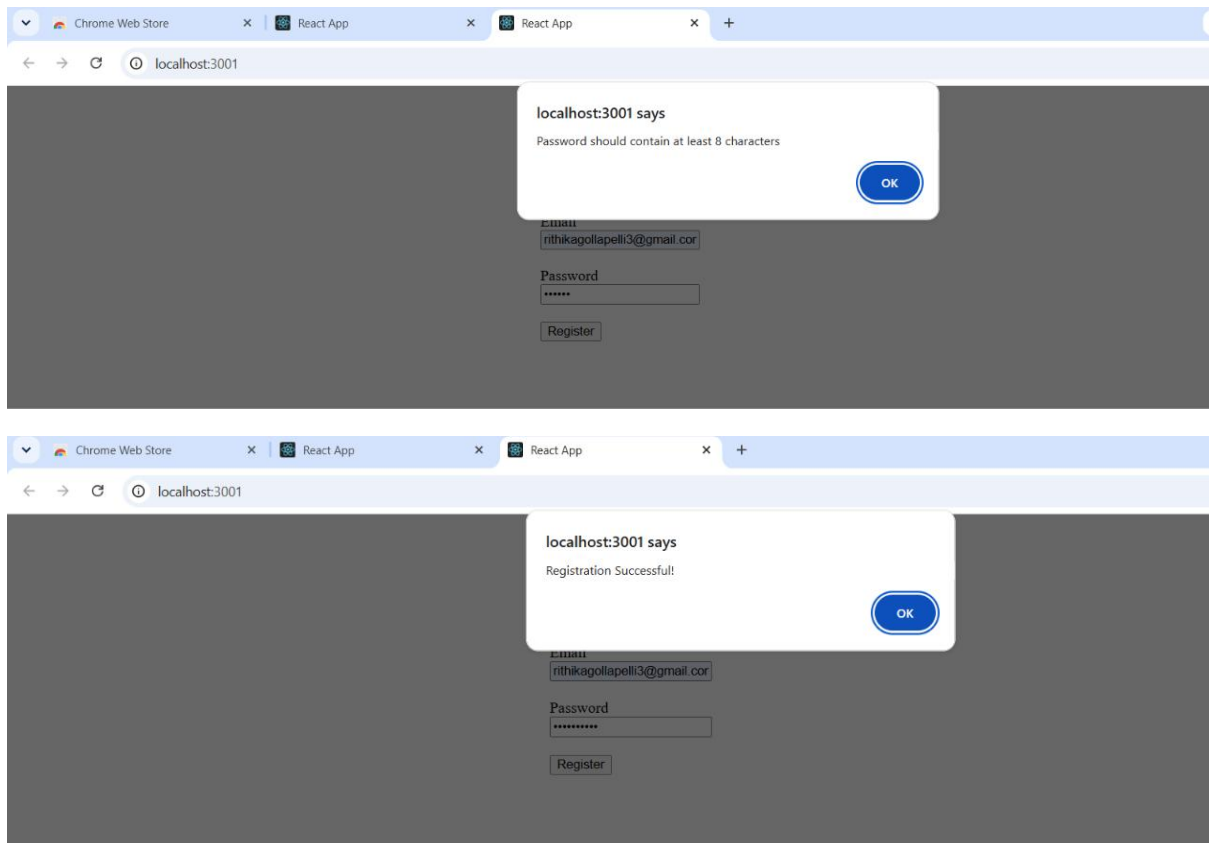
```
function Register() {  
  type="email"  
  value={email}  
  onChange={(e) => setEmail(e.target.value)}  
  />  
  <br /><br />  
  <label>Password</label>  
  <br />  
  <input  
    type="password"  
    value={password}  
    onChange={(e) => setPassword(e.target.value)}  
  />  
}
```

The terminal window at the bottom shows the following output:

```
Compiled successfully!  
  
You can now view exe16-mailregisterapp in the browser.  
  
Local:      http://localhost:3001  
On Your Network:  http://192.168.137.62:3001  
  
Note that the development build is not optimized.  
To create a production build, use npm run build.  
  
webpack compiled successfully
```

The page displays:





This implementation satisfies the HOL requirements for React form validation, controlled components, and validation through form event handlers and form submission.

Objectives

1. Explain React Forms Validation

React Forms Validation is the process of checking whether the user has entered valid and complete data before the form is submitted. Validation ensures that required fields are filled correctly and prevents invalid data from being processed or stored. It improves data accuracy, enhances security, and provides a better user experience.

Benefits

- Prevents invalid or incomplete data.
- Improves user experience.

2. Identify the Differences Between React Form and HTML Form

React Form

Uses React state to manage form data.

Uses controlled components.

Supports real-time validation.

Handles form submission using the onSubmit event.

HTML Form

Uses the browser to manage form data.

Uses native HTML form elements.

Validation is mostly handled after submission or through HTML attributes.

Submits data directly to the server using the action attribute.

React Form

Prevents page refresh using `event.preventDefault()`.

Suitable for Single Page Applications (SPAs). Commonly used in traditional web applications.

HTML Form

Reloads the page after form submission unless otherwise configured.

3. Explain About Controlled Components

A Controlled Component is a form element whose value is controlled by React state. Every change made by the user is immediately reflected in the component's state through the `onChange` event, making React the single source of truth for the form data.

Advantages

- Keeps form data synchronized with state.
- Simplifies validation.

4. Identify Various React Form Input Controls

React supports all standard HTML form controls for collecting user input.

Common Input Controls

Input Control Purpose

text	Accepts text input.
password	Accepts hidden password input.
email	Accepts email addresses.
number	Accepts numeric values.
checkbox	Allows multiple selections.
radio	Allows selecting one option from multiple choices.
select	Displays a dropdown list.
textarea	Accepts multiple lines of text.
date	Allows date selection.
file	Allows file uploads.

5. Explain How to Handle React Forms

Handling React forms involves storing user input in state and updating it using the `onChange` event. This ensures that the latest form values are always available within the component.

Benefits

- Enables real-time updates.
- Simplifies validation.